Entertainment Computing 38 (2021) 100412

Contents lists available at ScienceDirect

Entertainment Computing

journal homepage: www.elsevier.com/locate/entcom

The impact of video games on Students' educational outcomes

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ARTICLE INFO

Keywords:
Online gaming
Social networking
Education
Students
Saudi Arabia

ABSTRACT

The use of video games among both youths and elders has been recognised as a remarkable trend and a global success for the video game companies. Although some of these games might be useful in certain academic areas, the majority of students have been using them from a leisure perspective and consequently became addicted users. Video games are also considered a main constraint that hinder high learning outcomes amongst university students. Due to the dearth of any research correlating the influence if such games to students' learning outcomes in the Middle East in particular, we offer this study, which has two major objectives. First, it will explore whether gaming students achieve a different Grade Point Average (GPA) than their non-gaming peers or otherwise. Secondly, it will distinguish the GPA results of the intra-group students who are classified as heavy and non-heavy gamers. We adopted a quantitative approach and collected our data from a small public university situated in Saudi Arabia. Students with high academic achievement spend more time playing video games ($p = .005$), indeed almost as much as they spend on learning activities, and surprisingly they still earn high academic grades. The findings of this research, may improve the effectiveness of the strategies adopted by the online gaming industry to further improve these games with regard to education, as well as for behavioural and physiological studies to better reflect the interpretation of this phenomenon from a diverse perspective. A conclusion and the implications of this research are provided at the end of this study.

1. Introduction

The use of video games is, today, a cornerstone of both youths' and elders' lives. A video game is a game that is played through an electronic device. Past research has proposed a number of distinct advantages that video games can offer including, for instance, the educational perspective [1], social perspective [2], and financial returns for companies [3]. Video games can help students to gain academic learning through reducing their stress levels, contextualising game technical terms in the classroom content, as well as acting as a self-reward scheme. Beside the learning and educational activities, video games can also help youths, and students in particular, to gain further non-academic skills including, for instance, the ability to solve problems, work under pressure, make new friends through social networking, ability to cope with difficulties, and engaging in teamwork. Furthermore, video games help students to complete other activities including sharing information, chatting, paying for extra features for the game, learning, browsing and shopping, saving, and enhancing the language which the video game uses. Therefore, these video games have become increasingly integrated into students' everyday lives and consequently became a lifestyle not only for university students, but also for other students in other educational levels.

Updating the features of the video games on a regular basis could trigger loyalty to such games among users and therefore addiction in a later stage. For instance, Teng [4] conducted a study of 5144 online gamers, finding that certain features including, for instance, length depth, and breadth could be positively correlated to goal proximity - defined as the perception that a goal is close - which could be further positively correlated to personal motivation and online game loyalty. In a similar vein, Liao and Teng [5] conducted a study of 2025 online gamers, finding that the expectancy for growth is also positively related to perceived skill and perceived challenge, which will consequently positively influence online gamer loyalty. Equally, a study by Li et al. [6] of 1384 online gamers indicated that perceived avatar appearance, agreeableness and attractiveness are positively related to avatar friendliness, which in turn is positively related to online gamer loyalty.

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https://doi.org/10.1016/j.entcom.2021.100412
Received 15 January 2020; Received in revised form 27 January 2021; Accepted 28 January 2021
Available online 2 February 2021
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It is therefore vital to understand how this loyalty and addiction to video games affects students with regard to their educational outcomes so as to understand the correlation between the two streams. Therefore, we put forward two hypotheses for this study: (H1) the gaming students will achieve a different Grade Point Average (GPA) than their non-gaming peers; and (H2) students who are heavy gamers will achieve a different GPA than their non-heavy gamer peers. A Grade Point Average is the average of all grade points earned by the student during the studies. GPA is one of the most important indicators of college success [7], which can range from 0.0 (the lowest) to 4.0 (the highest). At the university, where the research was conducted, students are graded in letters. Each grade is assigned a point value, A = 4, B = 3, C = 2, D = 1, and F = 0. As not all exams are of the same difficulty, more difficult exams carry more credit hours. Therefore, the cumulative weighted by credit hour GPA is obtained by dividing the total grade points achieved by the total credits attempted:

$$GPA = \frac{\sum grade\ \ points}{\sum credit\ \ hours} = \frac{\sum(grade\ \ value \cdot credit\ \ hours)}{\sum credit\ \ hourss}$$

The example of two students (Table 1) with identical grades from different subjects shows an example of weighted GPA calculation. As we can see, the first student who gets a higher grade from the harder exam and a lower one from the easy one has a higher GPA than the student who got a lower grade from the harder exam and a higher grade from the easier one, 3.62 vs. 3.38. Obviously, the student A achieved a higher GPA.

2. Literature review on video games and Students' success

Mainstream media and the general public often have the view that playing video games is harmful and dangerous [8–10]. The authors of this paper are not gamers, but we have recollection of our childhood and how common it was to warn the youth that reading comics would make one stupid [11], or that Rock and Roll music was evil [12]. Nevertheless, video games, comics, and rock music were not the only dangerous entertainment activities for youth, according to the media. One could always find examples of the media being convinced that every new trend in entertainment represents a possible trigger for a crisis in society. The first 'media madness' about alleged danger directed towards teenagers can be found in 1866 when the UK publisher Edward Brett began publishing easy-reading crime stories that contained blood, violence, and crime [13]. These stories were targeted at children as he used schoolboys as the protagonists in his stories. The entirety of the Victorian society stood firmly against these publications. At same the time, crime was decreasing during the period of Brett's publications. The fear of motion pictures in the 1920s is another example of this 'media madness' [13]. The book *Our Movie Made Children*, written by Henry James Forman, with the exceptional assistance of media, rose to become a national best seller. The book contained questionable pseudo-scientific experiments supposedly demonstrating how movies posed a danger to the minds of children and teenagers.

Video games are a significant part of the lives of the majority of teenagers. In the USA, as much as 99% of boys and 94% of girls (overall 97% of teenagers) aged 12–17 play computer, web, portable, or console games [14]. According to the Entertainment Software Association, the video games market in 2015 was worth US$ 23.5 billion. However, video games are often viewed oversimplified by the general public and the mainstream media as violence-producing, isolating activities that waste children's time and leads to no positive outcome [15]. Financial returns are one of the main streams for video game developers. One of these recent famous examples is the live streaming video game named Player Unknown's Battlegrounds (PubG), which generated approximately eleven million USD within three days of its release and had sold one million copies within a month [16].

Playing video games is a significant part of many people's leisure time. One could argue that gamers prefer to play games in their leisure time and that they do it more often than those who are not as interested in such activities. Economics theory has a very clear and an unambiguous attitude about leisure time and gaming. Economic growth vanishes as the economy converges towards a steady state with a high ratio of leisure time and a constant level of production and consumption, respectively [17]. A study of young boys aged 6 and 9 [18] shows that once boys receive their first video game system they do not progress as quickly in school as boys who do not own such devices. Gamers have lower reading and writing scores and greater teacher-reported academic problems at follow-up than boys who were not involved in gaming activities. This represents clear evidence that gaming can lower the students' academic results, but on the other hand such a young group does not form a good basis for comparison with our mature students. Studies of adolescents aged 14.5 years [19] showed that an extra hour of daily screen time leads to two grades lower success in school in comparison to students aged 16. The authors recommended 1–2 h per day of entertainment screen time for juniors. A synthesis of research studies [20] showed a relationship between watching television and achievements. There is an optimum of 10 h per week of screen leisure time. If screen leisure time per week is under or over 10 h/w, academic achievements are lower. Adolescents achieve optimal academic performance when homework activities lasting 4 h/day [19]; participants at extreme ends of the scale, who do the least and the most reading and homework had lower academic performance. Obviously, gaming will not jeopardize academic results if students allocate the suggested time of 4 h/day to study; however, with 4 h of academic obligations, there is a timeframe that allows for leisure time and gaming, even during exam periods.

There is a cost of leisure, as economics teach us – for an hour of leisure that one decides to consume, one gives up an hour's wages [21]. However, IT and online technology could decrease the supposed cost of leisure time, as one research effort suggests [22]. Technology could help to manage leisure time in a productive and social manner. Participants successfully used their free time to enjoy themselves and, at the same time, study for their own personal and professional development. Another research effort by Willet & Gumulak [23] pointed out that entertainment and challenge were key reasons for playing video games. However, 89% of respondents answered that they had learned something new from gaming, including skills applicable in real-world situations. Current researchers in cognitive science reveal that good video games incorporate good learning principles, because if no-one could learn contemporary increasingly complex video games, then no one would buy them [24]. Video games incorporate some elements of information literacy, and in addition may challenge learners using a problem-solving approach [23]. Modern video games can set up a learning situation that leads to real understanding [25]. Language and learning in a games setting fits well with how the human mind is built to learn and think; schools sometimes do the opposite. Video games are, like literacy and computers, a new way to engage students in deep and engaged learning. Computer and video games can help children to perform better at school. However, the key issue [26] is that computer and video games can also help adults to rebuild education for the post-industrial, high-tech world by thinking about learning in a new way.

According to a nationwide survey in the United States [14], it is a

Table 1
A weighted GPA example.

Student	Course Code	Letter Grade	Grade Value	Credit Hours	Grade Points
A	303	A	4	5	20
	201	B	3	3	9
Total				*8*	*29*
GPA					**3.62**
B	303	B	3	5	15
	201	A	4	3	12
Total				*8*	*27*
GPA					**3.38**

stereotype that frequent game players are socially isolated loners. In fact, frequent gamers are just as communicative and socially engaged as less-active gamers, with one surprising difference: frequent gamers do send texts messages, but less frequently than less-active gamers at 32% vs. 41%. Games can influence social opportunities, teamwork, creativity, benefits for the brain, and problem-solving skills. Therefore a leading Midland UK headmaster thinks that he had managed to grasp the secret of exam success [27]: not reading long nights until you felt totally exhausted. In addition to studying for the exam, this headmaster also urged his students to sleep well, hang out with their friends and play games consoles (emphasis ours).

3. Methodology

The research was conducted at a small public university in Saudi Arabia. The aim of the paper is first to explore wither gaming students achieve a different GPA than their non-gaming peers, and secondly to distinguish the GPA results of the intra-group students who are classified as heavy and non-heavy gamers. Since this study is exploratory in nature, in focussing on investigating the gaming habits of senior year students, a quantitative approach was adopted. It is important to acknowledge that although the use of interviews would give more meaningful data, the authors preferred to distribute surveys due to possible restrictions with communicating in the English language with selected students as well as a possible lack of sincerity. Therefore, the students were approached through an online questionnaire developed through Google Forms. The online survey is reasonably inexpensive and saves time and effort [28]. The sampling method is graded as a critical case sampling [29] due to the fact that leisure activities are quite limited in Saudi Arabia due to the Sharia law restrictions in The Kingdom [30]. Therefore, the sample population and the sample itself were recognized as a critical case. Actually, in the Saudi environment, gamers have very few options how to spend their leisure time, and thus the Saudi gamers can sometimes be more devoted to video game playing than their Western peers.

Even with the questionnaire survey, such a large number of respondents was not anticipated. It was satisfying to receive 19 answers, for this number was approximately 20% of all senior students. Of those 19, two questionnaires were not valid and therefore excluded. So, the final number of valid surveys was 17 (n = 17) and whose answers are summarized in Table 2. There is one obvious limitation to this research. A sample of just 17 participants is critically small, but this was otherwise expected since we undertook this research at our very small university and we knew that we could not expect a normal distribution. However, even that sample could be suitable for advanced inferential statistical testing using nonparametric statistical tools. Accordingly, we applied the Mann-Whitney and Kruskal-Wallis tests. Data were analysed using IBM's SPSS Statistics for Windows, version 25, applying the Exact Tests to more accurately analyse of small and nonparametric datasets.

4. Data analysis and results

The participants were given an explanation of the purpose of the study and asked to complete the surveys. Primarily, the students were asked if they liked to play video games. Further on, we wanted to find out how much time, on average, in hours per day, they were spending on leisure activities and on study (preparing for exams) during the period leading up to their finals. We gave respondents the choice between five already stated leisure activities and one activity as an open (and unused) field. Those leisure activities were: Video games (PlayStation, XBOX, Mobile phones, PC); Non-video games (Cards, Monopoly, etc.); Hanging out with friends; Watching TV, movies, sitcoms, etc.; Surfing the Internet; Napping (extra sleep); and Other. The following question was the key issue: How much time, on average, in hours per day, did you spent on video games during the period leading up to the finals? Finally, after two more questions on maximum and minimum time spent time of playing video games, the next key question was: 'What GPA did you score in the finals?'

The two most popular leisure activities were internet surfing (almost 90% of students) and video games. Twelve students were active gamers or 70% of the sample, even during the exam period. Gamers were spending as much as three-quarters of their leisure time on gaming activities. All other leisure activities offered in the questionnaire were found in half of the answers of less. Five non-gamers spent 5.8 h on study and 1.5 h on leisure activities (Table 3). Gamers spent less on studying – 5 h, but considerably more on gaming and leisure, 2.54 and 3.33 h respectively (Fig. 1). Those data were related to students' success at studying. The first finding showed that non-gamers achieved better academic scores in comparison to gamers, with a GPA of 3.05 vs. one of 2.79, respectively.

In order to test the statistical significance of this finding, we compared the two groups (gamers vs. non-gamers) using the Mann-Whitney test (Table 4). The results of the test were not statistically significant: $U = 21.500, z = -.931, p = .352, r = -.226$. This test was used to determine the differences between two independent groups or categories, which in our study are called gamer and non-gamer students.

To reveal the optimum extent of video game playing, all students who played video games were assigned to one of three groups according to their GPA during their finals. The first group was the group of students with the weakest results, with a GPA of 2.00 – 2.49. They spent 4.63 h learning and 2.13 h playing video games. The second group spent more time on learning and less time on playing games, and achieved better

Table 2
Descriptive results relating to video games by respondent.

No.	Leisure in h	Studying in h	Average h on games	Do you like gaming	Max h on games	Min h on games	GPA	Avg GPA
1	6	**3**	**2**	YES	3–5 h	1 h	3.00–3.49	3.25
2	1	**5**	**0**	NO	–	–	3.00–3.49	3.25
3	0	**7**	**0**	NO	–	–	3.00–3.49	3.25
4	1	**5.5**	**1**	YES	1–3 h	Less than 1/4 of an hour	2.50–2.99	2.75
5	2.5	**6**	**3**	YES	1–3 h	Less than 3/4 of an hour	More than 3.50	3.75
6	5	**5**	**0**	NO	–	–	2.00–2.49	2.25
7	2	**2**	**4**	YES	5–7 h	2–3 h	2.00–2.49	2.25
8	1.5	**2.5**	**0.5**	YES	1–3 h	Less than 1/4 of an hour	2.00–2.49	2.25
9	2	**10**	**2**	YES	5–7 h	Less than 3/4 of an hour	2.50–2.99	2.75
10	0	**7**	**0**	NO	–	–	More than 3.50	3.75
11	7	**4**	**3**	YES	5–7 h	Less than 3/4 of an hour	2.00–2.49	2.25
12	1	**2**	**2**	YES	1–3 h	1 h	2.50–2.99	2.75
13	4	**8**	**3**	YES	–	Less than 1/4 of an hour	2.50–2.99	2.75
14	1.5	**5**	**0**	NO	–	–	2.50–2.99	2.75
15	4	**10**	**1**	YES	1–3 h	Less than 1/2 of an hour	2.00–2.49	2.25
16	2	**5**	**2**	YES	1–3 h	0	3.00–3.49	3.25
17	7	**2**	**7**	YES	5–7 h	More than 3 h	3.00–3.49	3.25
Avg	2.79	5.24	1.79					2.87

Table 3
Descriptive statistics.

Are you a gamer?	Number of students	Percentage	Average studying time	Average leisure time	Average gaming time	Average GPA
YES	12	70.59%	5.00 h	3.33 h	2.54 h	2.79
NO	5	29.41%	5.80 h	1.50 h	0.00 h	3.05

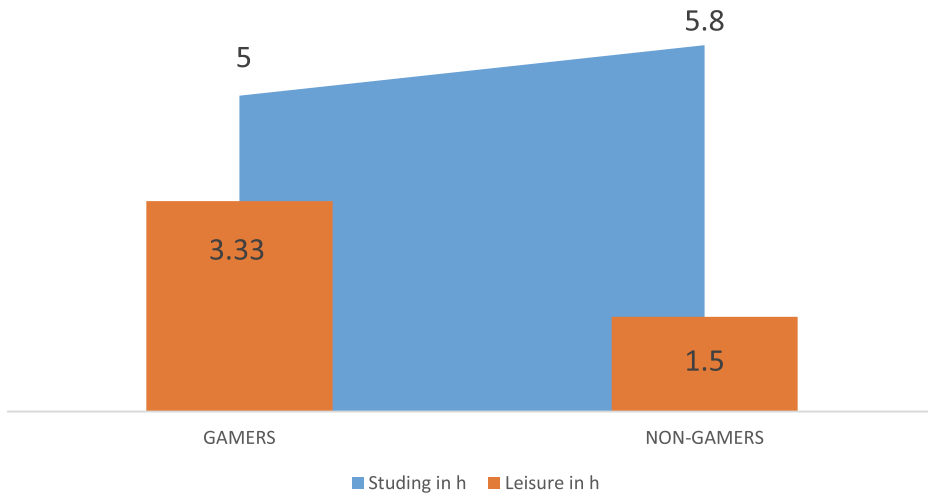

Fig. 1. Comparison of Gamers and Non-Gamers Study Time and Leisure time.

Table 4
Mann-Whitney test results.

Ranks		
N	Mean Rank	Sum of Ranks
12	8.29	99.50
5	10.70	53.50
17		

Test Statistics[a]	
	GPA
Mann-Whitney U	21.500
Wilcoxon W	99.500
Z	-0.931
Asymp. Sig. (2-tailed)	0.352
Exact Sig. [2*(1-tailed Sig.)]	.383[b]

[a] Grouping Variable: Gaming.
[b] Not corrected for ties.

results. In order to achieve a GPA in the range of 2.50 – 2.99, they spent 50% more time on learning and 33% less time on leisure activities compared to the first group. Having spent 2 h playing video games, members of this group spent the smallest amount of time playing. One might expect that the trend implied by the first two groups would continue with the last group, which consisted of the brightest students with a GPA of above 3. They spent 4 h on learning activities, even less than the weakest group, but in the line with the suggested time of learning for 4 h, as presented in our literature review [19]. On the other hand, the best group constituted the heaviest game players. They played more than all the other students, and indeed played above average. The average time they played video games playing during the exam period was 3.5 h per day, and yet they managed to achieve a GPA of between 3 and 4. For the participants of the third group, gaming was the most important activity in their leisure time. Gamers with the highest GPA were spending considerably more time on gaming activities during their leisure time in the exam period (Fig. 2) compared to the other gamer groups, at four-fifths versus three-quarters of their leisure time on games.

In order to test the statistical significance of the above, the Kruskal-Wallis test was applied (Table 5) in order to determine if there was a statistically significant difference between two or more of the groups or categories of measurement. The test indicates that there was a significant difference in the medians, $H(2, n = 17) = 10.748, p < .05$. Post-hoc tests were conducted to evaluate pairwise differences among the three groups, controlling for Type I error across tests by using the Holm's sequential Bonferroni approach. Mann–Whitney tests were used to follow up this finding, with all effects reported at a 0.0167 level of significance. The results of these tests indicated a significant difference in the medians between the groups. Between the groups with a low GPA and a high GPA, $z = -2.530, p < .0167, r = -.894$, between groups with a medium GPA and a high GPA, $z = -2.530, p < .0167, r = -.894$, and between groups with a low GPA and a medium GPA, $z = -2.646$, $p < .0167, r = -.936$.

This finding is also represented in the following figure (Fig. 3), which shows a relative comparison of gaming versus the GPA achieved by each of the two groups of selected students. The difference is clearly noticeable, and therefore reflects the importance of time spent on gaming and how this apparent waste in time can negatively affect the educational performance of students at universities.

5. Discussion

The first finding shows that non-gamers ($Mdn = 3.25, n = 5$) achieved a better academic score in comparison to gamers ($Mdn = 2.75, n = 12$). Their GPA of 3.05 was higher than gamers' GPA of 2.79. Although students who did not play games achieved a 10% better GPA than their gamer peers, a Mann-Whitney test revealed no statistical difference in academic score between them, $U = 21.500, z = -.931, p = .352, r = -.226$. Therefore, the null hypothesis may not be rejected. The calculated effect size r represents a small to medium effect. Accordingly, we reject hypothesis H1 and conclude that there are no differences in GPA between students who are constant gamers and students who prefer not to play games during finals in our sample.

In order to test H2, all students who played video games were grouped into three different groups according to their GPA during finals. The first group of students with the weakest results ($Mdn = 2.25, n = 4$) spent 4.63 h learning and 2.13 h playing video games. The second group

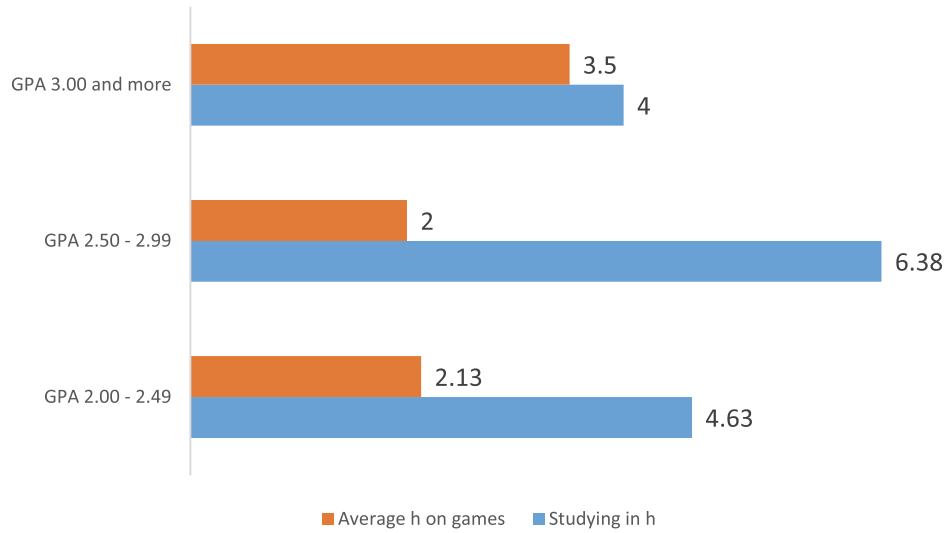

Fig. 2. Time Spent Studying and Time Spent Gaming vs. Gamers' GPAs.

Table 5
Kruskal-Wallis Test Results.

Ranks			
	Rang_GPA	N	Mean Rank
Avg_GPA	1.00	4	2.50
	2.00	4	6.50
	3.00	4	10.50
	Total	12	

Test Statistics[a,b]	
	Avg_GPA
Kruskal-Wallis H	10.748
df	2
Asymp. Sig.	0.005

[a] . Kruskal Wallis Test
[b] . Grouping Variable: Rang_GPA

($Mdn = 2.75, n = 4$) spent 50% more time on learning, and 33% less time on playing games, and achieved better results. The surprise came with the third group ($Mdn = 3.25, n = 4$). They spent even less time than the weakest group on learning activities, but they played more games than all the other students. Playing an astonishing 3.5 h video games daily during the exam period, they nevertheless managed to achieve the highest GPA result. For the participants of the third group, gaming was the most important activity in their leisure time. A Kruskal-Wallis test was conducted to evaluate the differences between these three groups of students. The test was significant, $H(2, n = 17) = 10.748, p < .05$. The third group (GPA 3.00 – 4.00) recorded a higher median score of 3.25 than the other two groups. The median for the second group (GPA 2.50 – 2.99) was 2.75, whilst median for the first group (GPA 2.50 – 2.99) was 2.25. Follow-up Mann-Whitney tests were conducted to evaluate pairwise differences between the three groups. The results of these tests indicated the significant differences between the groups, i.e., low GPA and high GPA, $z = -2.530, p < .0167, r = -.894$, mid GPA and high GPA, $z = -2.530, p < .0167, r = -.894$, and low GPA and mid GPA, $z = -2.646, p < .0167, r = -.936$. The effect size r in all three tests was much higher than 0.5, which indicates a significant effect. Accordingly, we accept hypothesis H2 and conclude that excellent students who are heavy gamers differ significantly from their non-heavy gamer peers with less good grades.

The findings of the study lead us to understand how to distinguish the working and gaming habits of excellent students compared to others. The above-average students are devoted to studying and to good grades. They only perceive effective study time as time spent on study, and not in preparation. Less successful students consider the entire time allocated to work to be study time, from the first to the last minute,

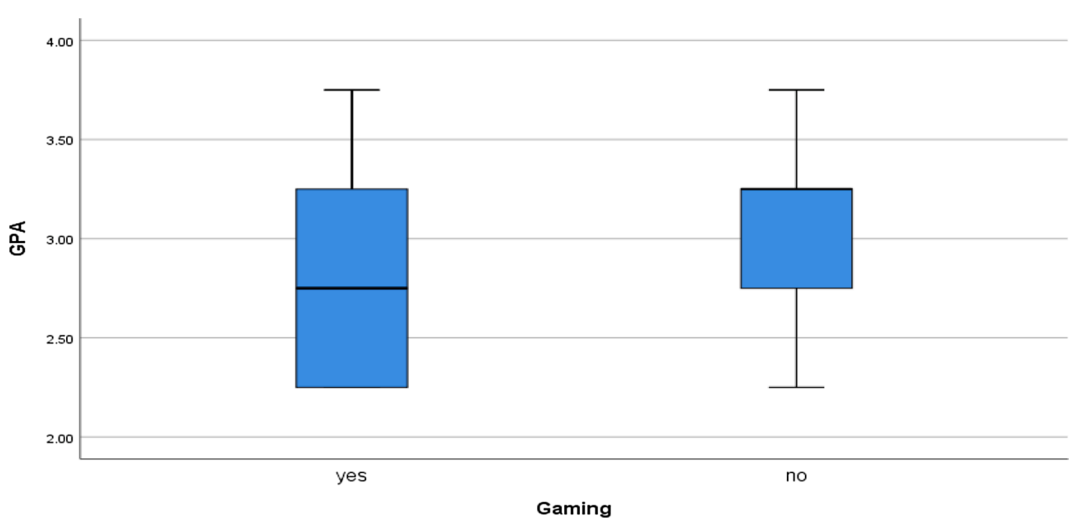

Fig. 3. Graphical Representation of Gaming versus GPA Achieved for the Two Groups.

including all short and, indeed, longer break. Clever students do not have the habit of taking a lot of short brakes. Their break is the pause required to mentally relax and recover their energy. Gaming is a highly suitable activity for both – relaxing and recuperating – and therefore excellent students can simultaneously be heavy gamers as well as heavy learners. It is not our intention to advocate heavy gaming habits, however, the message for good students would be, "Plan your study time as well as your leisure time and make the most of it. Gaming can help you utilize your leisure time in the most effective way possible, if you enjoy playing games. Do not worry about gaming, it is not a danger to you at all. During finals, as well, it could indeed help".

6. Conclusions and implications

With regard to hypothesis H1 (The gaming students will achieve a different GPA to their non-gaming peers), we cannot argue that playing video games will lower the study results of regular students. In our sample, the differences in GPA between the students who were gamers and those who were not was not significant. For hypothesis H2 (The students who are the heavy gamers will achieve a different GPA to their non-heavy gamers peers) it can be argued that a group of highly efficient and successful students, who know how to use time for study as well as gaming in an effective manner, do in fact exist. The excellent students who are the heavy gamers differ significantly from their non-heavy gamer peers. This group spent 3.5 h on average on video game activities, as well as 4 h on average studying. Therefore, with regard to the best students, the optimum level for gaming is more than 3 h and less than 4 h. This group was named the 'play hard – study hard' group. Average senior students spend an average amount of time playing video games. It is better for this particular group not to play games during their exam period, as they will score better results. Excellent students spend more time on video games, almost as much as they spend on learning activities, but can still earn high grades. Their formula for success is 'Play hard – study hard'.

What is the main the contribution of this research? We look at this paper and this research as an anecdotal case. However, although anecdotal, we also consider this case to be a critical one. Saudi Arabia at that time had (and indeed still has) a very unique position. The (male) students in a small desert city had no possibilities with regard to other leisure activities except those given in our questionnaire: gaming, controlled Internet, and censored movies. No coffee or sweet shops, one mall just for families, a hot climate and sand everywhere were not conducive to hanging out with friends. Moreover, spending time with a girlfriend or boyfriend is prohibited. Therefore, video gaming, which is extremely popular in Saudi Arabia, probably more than anywhere else, is almost certainly the only realistic leisure option for our students. Therefore, in light of this research, the unexpected finding that our best students were not distracted from gaming activities even during a tough exam period, but rather quite the opposite, is the most important result obtained from this research. Decision makers at local universities need to invest more in upgrading students' campus facilities to provide their students with the various activities that may help them both physically and mentally, instead of narrowing their leisure options which they in any case mostly use for gaming activities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.entcom.2021.100412.

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